

RESEARCH INTEREST

Applied Math and Statistics, AI Explanation, Bioinformatics, Sequence Alignment, Genomics and Proteomics Analysis, AI-driven for Medical Imaging, Quant Finance, Large language Models (LLMs) in Biomedical Research, Systems Biology and Precision Medicine, Medical Imaging.

WORK AND RESEARCH EXPERIENCE

- **Research Assistant Project: "Causal Narrative in Information Systems Research"** USA
Researcher Collaboration Jul 2025–now
Minh Nguyen (Florida Atlantic University) and Mike Nguyen (University of Southern California)

- **Research Material Documentation:** Collected and organized research documents and papers for literature review; managed references and maintained clear documentation to support the project workflow.
- **Data Collection Support:** Gathered casual narratives from surveys, interviews, and online platforms; logged sources, and key metadata to ensure traceability and research readiness.
- **Feature Preparation:** Transformed raw narratives into structured datasets by deriving interpretable indicators; documented workflows and produced clear summaries to support analysis and manuscript development.

- **Bioinformatics Research Intern: Entobel**

Researcher Intern May 2026–Present

- **Research Project:** *Development of a Nextflow Pipeline and Report Template for BSF Genomic Diversity and Inbreeding Monitoring*
- **Pipeline Development:** Designed and implemented a scalable **Nextflow-based bioinformatics workflow** for automated processing and analysis of Black Soldier Fly (BSF) genomic sequencing data.
- **Genomic Diversity and Inbreeding Analysis:** Developed analytical modules for genomic diversity assessment, population structure evaluation, and inbreeding coefficient monitoring across BSF breeding populations.
- **Workflow Automation and Reporting:** Integrated quality control, variant calling, statistical summarization, and automated report generation into a reproducible computational framework for genomic monitoring.
- **Industrial Bioinformatics Collaboration:** Collaborated with research and technical teams to optimize computational reproducibility and support data-driven breeding strategy decisions for sustainable industrial insect production.

- **Research Assistant: AI VIET NAM**

Researcher Collaboration May 2025–Jan 2026

- **To develop research ideas:** Concept notes, research proposals, and methodological design for AI-driven research and consultancy projects.
- **Implementing research and consultancy projects:** Designing study protocols, collecting and analyzing data, producing reports, and disseminating project results.
- **Team Collaboration:** Contributed to developing semi-supervised deep learning models for histopathology image segmentation, leveraging limited annotated data with consistency-based learning and feature regularization techniques. Enhanced tumor boundary detection and improved tissue classification performance in medical imaging workflows.

- **Department of Artificial Intelligence Convergence, Chonnam National University** South Korea
Research Assistant – BIEL Lab, Chonnam National University Hwasun Hospital 2021 – 2023

Research Portfolio (Slides, PDF)

- **AI-Driven for Medical Imaging:** Leveraging advanced deep learning models and computer vision techniques to detect and localize tumors in critical anatomical regions, from medical imaging data.
- **Research Project:**
1. Project Tumor Detection in Liver Imaging: Leveraging high-resolution with medical imaging as CT, MRI, 18F-FDG PET, and 11C-ACT PET scans to accurately identify and differentiate liver tumors, such as hepatocellular carcinoma (HCC) and metastases, with high precision. The models are trained for segmentation and object detection, capturing subtle tissue abnormalities and detecting cancer.

2. *Tumor Detection in Bone Imaging*: Applying region segmentation algorithms, such as Thresholding, Region Growing, Deep Learning-Based Segmentation (U-Net, Mask R-CNN) to detect bone lesions and metastatic tumors in skeletal structures. By analyzing MRI and PET scans, the models assist in identifying bone metastases, a common site for the spread of cancers, enabling early intervention and treatment optimization.

3. *Multimodal Imaging for Comprehensive Tumor Characterization*: Integrating data from 18F-FDG PET and 11C-ACT PET scans with CT and MRI to improve the accuracy and sensitivity of tumor detection. PET imaging provides metabolic activity information, enhancing the ability to differentiate malignant from benign lesions.

4. *Developed a deep learning-based model for analyzing Optical Coherence Tomography (OCT) images*: Combining speckle variance techniques and generated images (GANs model) to monitor the coagulation process in biological tissues.

- **Collaboration with Healthcare Professionals**: Worked closely with physicians and research teams to deepen understanding of anatomical knowledge and clinical challenges, ultimately developing automated prediction models for diagnostic purposes.
- **Conducting comprehensive literature reviews**: To understanding the existing research and advancements in deep learning application for medical purposes.

- **PropellerAds** Vietnam
Machine Learning Engineering 2024 – now
- **Fintech Company** Vietnam
Machine Learning Engineering and Data Mining
 - **Developing credit scoring model**: Utilized advanced statistical techniques, regression analysis, and machine learning algorithms to extract actionable insights from financial data and data services, optimizing decision-making processes and improving predictive accuracy in areas such as credit scoring, risk assessment, and financial forecasting.
 - **Risk Reduction in Credit Portfolio**: Reduced the percentage of customers with overdue payments by across the total portfolio by optimizing credit scoring models, improving feature selection, and enhancing predictive accuracy for risk management.

EDUCATION AND COURSES

- **Chonnam National University - Department of AI Convergence** Gwangju, South Korea
Master of Science 2021 - 2023
- **Banking Academy of Vietnam - Faculty of Finance** Hanoi, Vietnam
Bachelor of Arts
- **Johns Hopkins University - Department of Mathematics**
Real Analysis (MATH 110.405) Synchronous course
- **Louisiana State University - Department of Mathematics**
Linear Algebra (MATH 2085) Asynchronous course
- **CASED Center**
Optimization Offline course
- **Johns Hopkins University - Department of Nursing**
BIostatistics (NR.110.202) Fall 2026 ; Synchronous course

Scholarship Awardee: **Illinois Institute of Technology**

M.S. Applied Mathematics (Admitted with Merit Scholarship) [Proof] Fall 2026

- **Graduate Excellence**: Selected for admission based on strong quantitative, computational, and interdisciplinary research achievements.
- **Merit Recognition**: Awarded competitive merit-based funding for academic excellence and research potential.

CONFERENCES ATTENDED

- **Oral presentations**
 - Annual Meeting of ARCCNM-KSNM: International Conference
V.T. Nguyen, “Coagulation observation monitoring using deep learning Speckle variance optical coherence tomography”, South Korea
- **Present presentations**
 - The Korean Society of Medical & Biological Engineering
V.T. Nguyen, “Feasibility study to utilize GAN deep learning in blood coagulation monitoring with speckle variance optical coherence tomography”, South Korea

SKILLS

- **Programming languages:** Python, L^AT_EX, R, Matlab, HTML, CSS, C++
- **Tools, Framework:** SQL, Airflow, Pyspark, AWS Lambda, Pytorch, Git, Hugging Face
- **Softwares:** 3D-Slicer, SolidWorks, ITK-SNAP, MONAI
- **Sequence Alignment:** BLAST, Bowtie2, Minimap2
- **Transcriptomics Tools (RNA-Seq, scRNA-Seq):** DESeq2, Monocle, miRDeep, miRBase